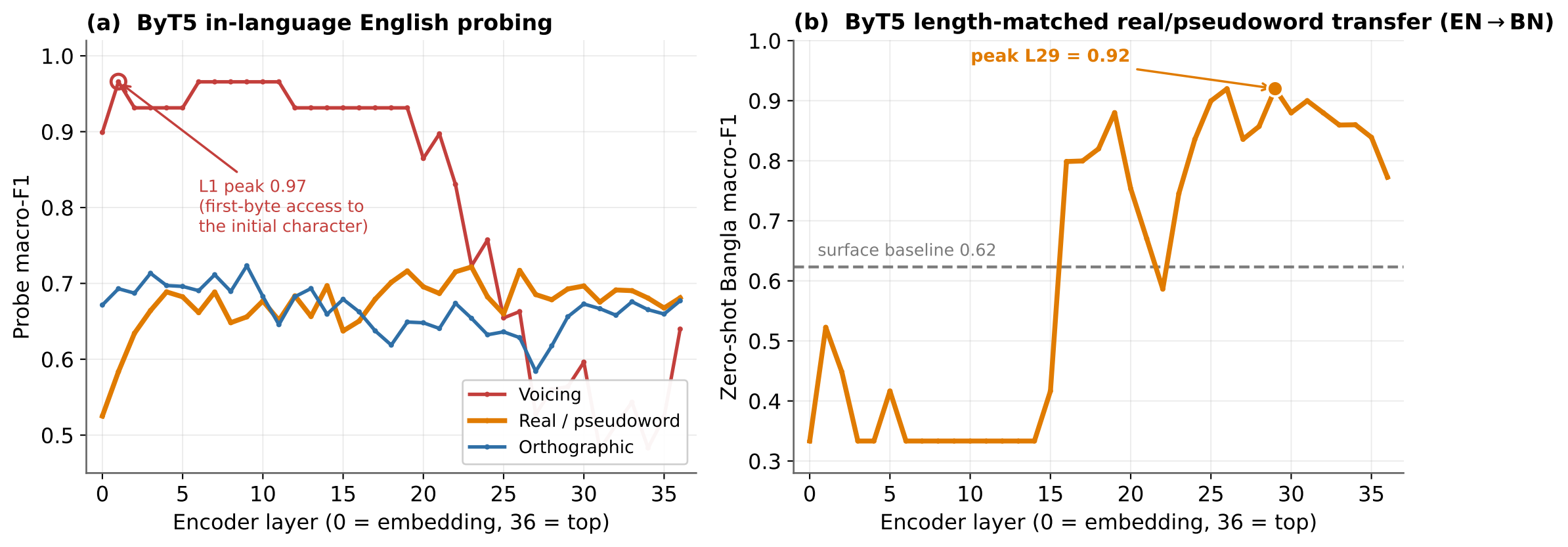
(a) ByT5 in-language English probing
Probe macro-F1
1.0
0.9
0.8
0.7
0.6
0.5
L1 peak 0.97
(first-byte access to
the initial character)
Voicing
Real / pseudoword
Orthographic
0
5
10
15
20
25
30
35
Encoder layer (0 = embedding, 36 = top)
(b) ByT5 length-matched real/pseudoword transfer (EN → BN)
Zero-shot Bangla macro-F1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
peak L29 = 0.92
surface baseline 0.62
0
5
10
15
20
25
30
35
Encoder layer (0 = embedding, 36 = top)